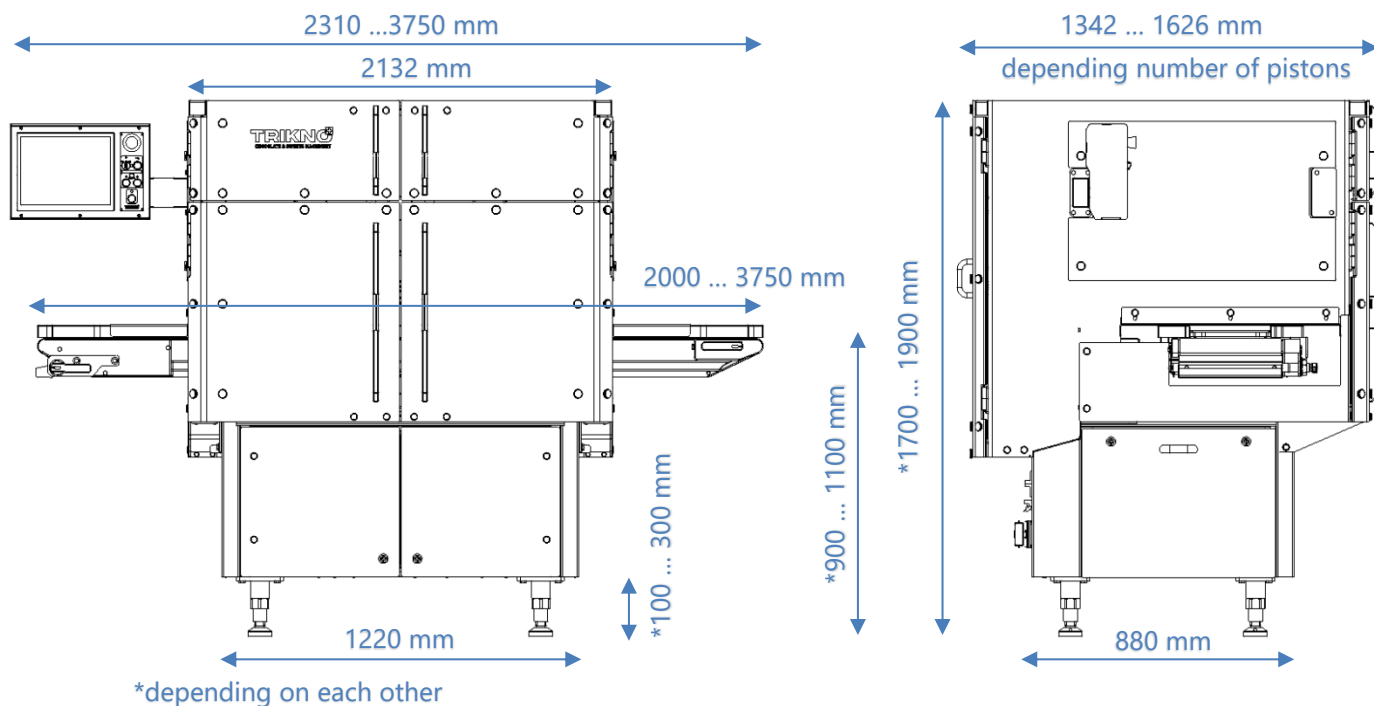
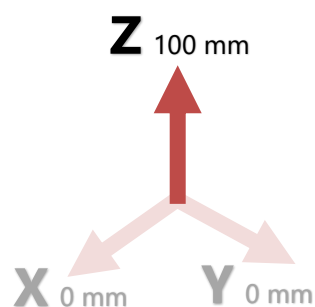
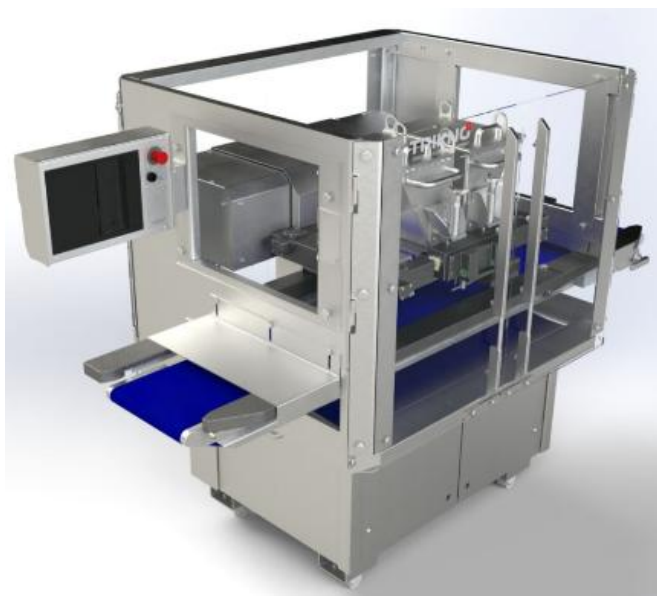




Technical Data Sheet TCM Alpha Compact

Machine with proven single-rotary piston system, powered entirely by electric servo motors.

Wide range of applications.



Single Boxer	9 9/18	12 12/24	18 18/36	24 24/48	Optional
Bushing width	35 mm		145 mm		
Piston distance	35 mm		45 mm / 22.5 mm		
Stroke	52 mm		60 mm		
Mould width	261 ... 412 mm				1261 ... 517 mm
Weight	1'250 kg	1'300 kg	1'450 kg	1'500 kg	

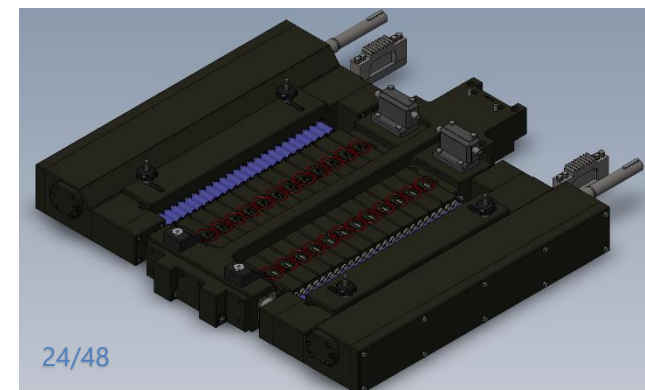
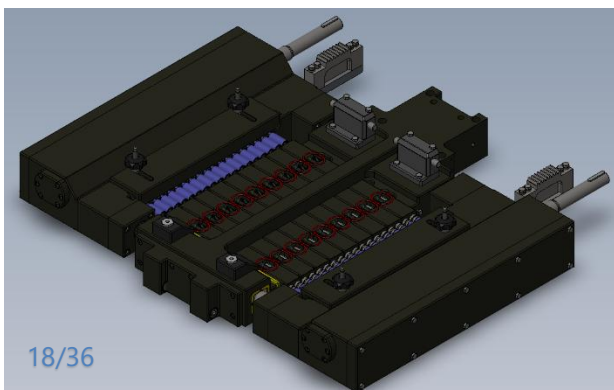
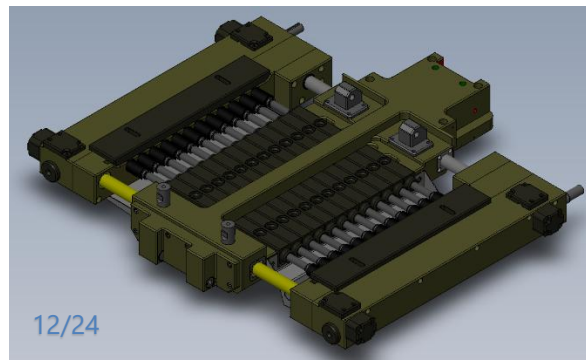
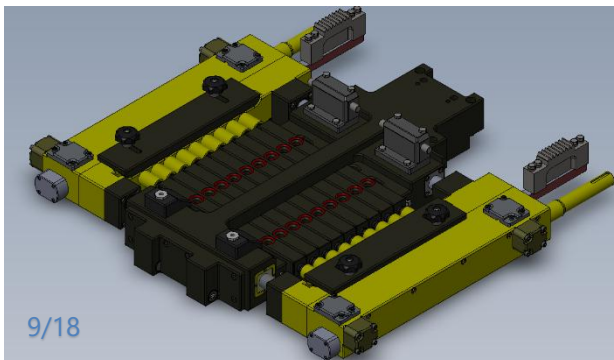


Characteristics

- ✓ Head movement in Z-axis 100 mm with electric servo motors
- ✓ High-precision single rotary piston system with identical dosing paths (proven for more than 30 years – Single Piston System)
- ✓ No mixing between the pistons possible (1 piston = 1 nozzle)
- ✓ Dosing machine designed for the confectioner or for laboratory tests
- ✓ Quick changeover of different casting tools (pistons/bushings/nozzle plates) without additional tools
- ✓ Number of pistons individually selectable (easy activation / deactivation by using individual couplings)
- ✓ Easy cleaning of the dosing machine without additional tools (Stainless steel or aluminium in ALTEF® coating – for pH values 4 ... 8)
- ✓ Easy cleaning of the pistons and bushings (no tools necessary)
- ✓ Touch-screen HMI for machine operation
- ✓ Casting head in free-arm design for easy operation / cleaning / adjustment
- ✓ Fully digital machine control
- ✓ Remote maintenance on PLC, HMI, servo drives possible
- ✓ Software designed for maximum accuracy and flexibility
- ✓ One Shot software already installed on every machine
- ✓ Double-walled hopper for chocolate side

Options

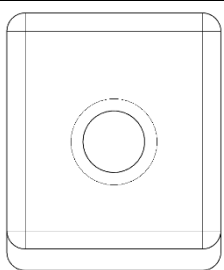
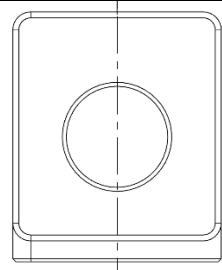
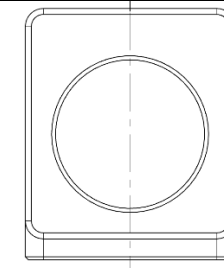
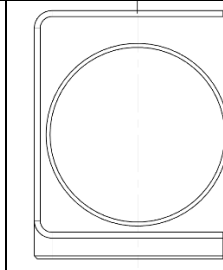
- Double-walled hopper also for filling side
- Piston lubrication system for one or both piston sides (drip lubrication)
- Linear agitator for one or both hoppers
- Triple Shot (1 filling / 2 shells OR 2 fillings / 1 shell)
- Curtain nozzle for filled bars or bars



Variants with 9 or 12 pistons per side

- Operating variants
 - Single (for drops to bars without filling)
 - Boxer (for One Shot chocolates, bars or blocks)
- Head variants
 - 9 pistons per side
 - 12 pistons per side
- Hopper Graduation Chocolate Side and/or Filling Side
- Belt running direction from left to right or right to left
- Pistons/bushings pairs in up to two different sizes

Piston Systems

Bushing width 35 mm Stroke 52 mm				
Piston- Ø	10 mm	17 mm	25 mm	30 mm
Volume	4.1 ml	11.8 ml	25.5 ml	36.8 ml
	4.9 g	14.2 g	30.6 g	44.1 g
Recommendation	Lids or drops	Pralines in One Shot or Triple Shot	Bars with or without small inclusions	Bars with or without small inclusions

Machine Capacities (examples with Boxer 9/18)

Example 1: 100g Bars without filling (solid)

Maximum design with two chocolate hoppers (3 nozzles per bar)

Each stroke 6 bars, 17 strokes/min and 60min/h results in:

- **6'120 bars/h or 612 kg/h**

Example 2: 10g One Shot Pralines

Maximum design with two hoppers (chocolate and filling)

Each stroke 9 pralines, 20 strokes/min and 60min/h results in:

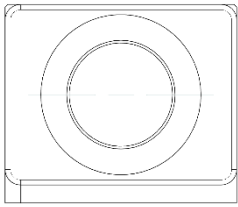
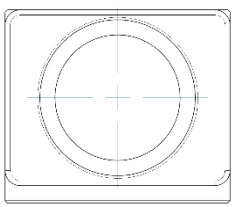
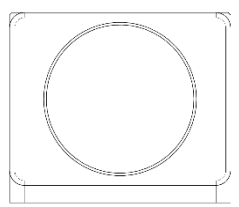
- **10'800 pralines/h or 108 kg/h**



Variants with 9 or 12 pistons per side

- Operating variants
 - Single (for drops to bars without filling)
 - Boxer (for One Shot chocolates, bars or blocks)
- Head variants
 - 9 pistons per side
 - 12 pistons per side
- Band width depending on mould width
 - For narrow moulds width from 261 ... 412 mm
 - For wide mould width from 366 ... 517 mm
- Hopper Graduation Chocolate Side and/or Filling Side
- Belt running direction from left to right or right to left
- Pistons/bushings pairs in up to two different sizes

Piston Systems

Bushing width 45 mm Stroke 60 mm				
Piston - Ø	10 mm	20 mm	25 mm	30 mm
Volume	10.6 ml	18.9 ml	29.5 ml	42.4 ml
	12.7 g	22.6 g	35.3 g	50.9 g
Recommendations	Lids or drops	Pralines in One Shot or Triple Shot	Bars with or without small inclusions	Bars with or without small inclusions

Machine Capacities (Examples with 12 pistons per side)

Example 1: 100g bars without filling (solid)

Maximum design with two chocolate hoppers (2 nozzles for each bar)

Each stroke 12 bars, 17 strokes/min and 60min/h results in:

- **12'240 bars/h or 1'224 kg/h**

Example 2: 10g One Shot Pralines

Maximum design with two hoppers (chocolate and filling)

Each stroke 12 pralines, 20 strokes/min and 60min/h results in:

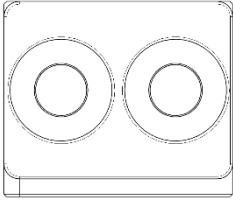
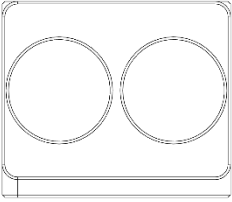
- **14'400 pralines/h or 144 kg/h**



Variants with 18 or 24 pistons per side

- Operating variants
 - Single (for drops to bars without filling)
 - Boxer (for One Shot chocolates, bars or blocks)
- Head variants
 - 9 pistons per side
 - 12 pistons per side
- Band width depending on mould width
 - For narrow moulds width from 261 ... 412 mm
 - For wide mould width from 366 ... 517 mm
- Hopper Graduation Chocolate Side and/or Filling Side
- Belt running direction from left to right or right to left
- Pistons/bushings pairs in up to two different sizes

Piston Systems

Bushing width 45 mm Stroke 60 mm		
Piston - Ø	10 mm	20 mm
Volume	4.7 ml	18.9 ml
	5.7 g	22.6 g
Recommendations	Lids or drops	Pralines in One Shot or Triple Shot

Machine Capacities (Example with 24 pistons per side)

Example 1: 100g bars without filling (solid)

Maximum design with two chocolate hoppers (6 pistons for one bar)
 Each stroke 4 bars, 17 strokes/min and 60min/h results in:

- **4'080 bars/h or 408 kg/h**

Example 2: 10g One Shot Pralines

Maximum design with two hoppers (chocolate and filling)
 Each stroke 24 pralines, 20 strokes/min and 60min/h results in:

- **28'800 pralines/h or 288 kg/h**



Applications



- Bars without filling
- Bars with small inclusions up to 3 mm



- Pralines filled in 2 colours (One Shot)
- Chocolate Bar 2-Colour Filled (One Shot)
- Pralines filled in 3 colours (Triple Shot)
- Chocolate bars filled (One Shot Curtain)



- Fillings for Macarons
- Fillings for Alfajores

